Levi-quotient descent and fibrewise flat connections for Cartan geometries on torus fibrations

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Editorial note. Original author mathematical text and the complete proof of the independent torus-fibration proposition. This packet is not a new count of the canonical developing-subvariety classification907.

Editorial note. The article explicitly identifies this torus-fibration result as independent of the later flat classification. Named external BG connection theory remains a prerequisite at original author granularity.

Editorial scope. the counted statement and its own proof environment only

Allowed standard prerequisites (editorial).

- Holomorphic principal bundles over complex tori: Biswas-Gomez Theorem4.1. author Lemma lemma:BG1274-1298 and complete original bibliography. Unique flat connection for the connected unipotent reduction; proof is not AI-expanded

Take a complex Lie group G and some finite-dimensional holomorphic G-modules V_i. A *reductive* (*semisimple*) *Levi quotient* of G over $\{V_i\}$ is a quotient of complex Lie groups

$$1 \longrightarrow G_{\mathrm{u}} \longrightarrow G \longrightarrow G_{\mathrm{r}} \longrightarrow 1,$$

and a complex Lie group morphism $G_{\mathrm{u}} \longrightarrow \overline{G}_{\mathrm{u}}$ to a unipotent (solvable) complex linear algebraic group, injective on Lie algebras, so that G_{r} is a reductive (semisimple) complex linear algebraic group and so that the action of G_{u} on each V_i factors through a polynomial action of $\overline{G}_{\mathrm{u}}$. It follows that G_{u} preserves a filtration of G-modules on each V_i by Lie's theorem (see [55, Theorem 3]), and acts trivially on the quotients V_{i+1}/V_i [9, Section 4.8].

A complex homogeneous space $X = G/H$ is *infinitesimally algebraic* if, after perhaps replacing G and H by finite index subgroups, H has a reductive Levi quotient over $\mathfrak{g}$, for which the action of H_{r} on $\mathfrak{g}/\mathfrak{h}$ is faithful, and G has a semisimple Levi quotient over $\mathfrak{g}$.

12 Bundles on tori

In this section, we consider how to relate holomorphic connections on different principal bundles. We need this because the standard theory of the Shafarevich

fibration (see the standard reference [63]) is not developed for general principal bundles, but only those with particular structure groups.

Lemma 12.1 ([5, Theorem 4.1])**.** *Take a holomorphic principal bundle $G \longrightarrow P \longrightarrow T$ on a complex torus T, with structure group G. Suppose that some finite index subgroup $G_0 \subset G$ has semisimple Levi quotient $1 \longrightarrow G_{\mathrm{s}} \longrightarrow G_0 \longrightarrow G_{\mathrm{ss}} \longrightarrow 1$. The following are equivalent:*

(1) *the bundle admits a holomorphic connection,*

(2) *the bundle is pseudostable with vanishing first and second Chern classes,*

(3) *after perhaps lifting to a finite unramified covering torus, the associated bundle G_{ss} admits a holomorphic reduction of structure group to the Borel subgroup of G_{ss}, so that the line bundle associated to any character of the Borel subgroup has vanishing first Chern class,*

(4)
- *the bundle is pseudostable,*
- *the bundle admits a unique holomorphic flat connection,*
- *every holomorphic section of any associated vector bundle is parallel for that holomorphic flat connection,*
- *every holomorphic connection on the bundle is parallel in that holomorphic flat connection.*

13 Cartan connections on torus fibrations

In this section, we extend our theory of principal bundles on tori to principal bundles on torus fibrations as they occur in Cartan geometries.

Proposition 13.1. *Take a holomorphic torus fibration $M \longrightarrow S$ over a complex manifold S. Take a holomorphic Cartan geometry $H \longrightarrow E \longrightarrow M$ with infinitesimally algebraic model (X, G). Then the quotient bundle $H_{\mathrm{r}} \longrightarrow E/H_{\mathrm{u}} \longrightarrow M$ is the pullback of a holomorphic principal bundle $H_{\mathrm{r}} \longrightarrow E_{\mathrm{r}} \longrightarrow S$. On each fiber of $M \longrightarrow S$, the bundle E pulls back to have a holomorphic flat connection with holonomy in H_{u}, and this connection varies holomorphically with the fiber.*

Proof. Take an H-invariant filtration $0 = F_0 \subset F_1 \subset \cdots \subset F_N = \mathfrak{g}/\mathfrak{h}$ so that H_{u} acts trivially on the associated graded $W = \mathrm{gr}_* F$. Then $Z := \mathrm{GL}(W)/H = \mathrm{GL}(W)/H_{\mathrm{r}}$ is a quasi-projective homogeneous variety [59, Theorem 12.2.1], and indeed affine; see Mumford et al. [50, Theorem 1.1] and Procesi [53, Theorem 2].

Denote the fibration $s\colon M \longrightarrow S$. For each $s_0 \in S$, let $M_{s_0} := s^{-1}\{s_0\}$. For each point $s_0 \in S$, let V_{s_0} be the vector space of translations of the fiber M_{s_0}, and let $\Lambda_{s_0} \subset V_{s_0}$ be the discrete subgroup of translations acting trivially on M_{s_0}, making a vector bundle $V \longrightarrow S$ and a covering space $\Lambda \longrightarrow S$ with $\Lambda \subset V$, so that $M = V/\Lambda$.

Since the problem is local on S, we can assume that there is a holomorphic section $m\colon S \longrightarrow M$ of the bundle $s\colon M \longrightarrow S$. We can assume that S is an open subset of $\mathbb{C}^n$ with local holomorphic coordinates z^i, and take local sections $e_1, e_2, \ldots, e_p$ of $\Lambda \longrightarrow S$ so that they form a basis of local sections of V. Write any element of the total space of V as $w = w^A e_A$. On the total space of V, we

have local holomorphic coordinates z^i, w^A. Since $M = V/\Lambda$ is covered by V, these are also local holomorphic coordinates on M. Pull back the 1-forms $\mathrm{d}z^i$, $\mathrm{d}w^A$ to E, where they are a basis of the semibasic 1-forms. The 1-form $\omega + \mathfrak{h} \in \Omega^1_E \otimes^H (\mathfrak{g}/\mathfrak{h})$ is semibasic, so $\omega + \mathfrak{h} = f_i \mathrm{d}z^i + f_A \mathrm{d}w^A$ for unique holomorphic functions $f_i, f_A \colon E \longrightarrow \mathfrak{g}/\mathfrak{h}$. Since ω is a linear isomorphism on each tangent space of E, $\omega + \mathfrak{h}$ is an isomorphism of the pullback tangent bundle to $\mathfrak{g}/\mathfrak{h}$, i.e., if we identify the 1-forms $\mathrm{d}z^i$, $\mathrm{d}w^A$ with $\mathfrak{g}/\mathfrak{h}$ by any fixed linear isomorphism, then

$$f := (f_i \ f_A) \colon \ E \longrightarrow \mathrm{GL}(\mathfrak{g}/\mathfrak{h})$$

is an H-equivariant holomorphic map $r_h^* f = \mathrm{Ad}_h^{-1} \circ f$, $h \in H$. Hence the quotient object

$$\overline{f} \colon \ M = E/H \longrightarrow Z = \mathrm{GL}(W)/H_{\mathrm{r}}$$

is a holomorphic map to an affine variety. So $\overline{f}$ is constant on each torus fiber, i.e., drops to a map $\overline{f} \colon \ S \longrightarrow Z$ giving maps

$$\begin{array}{ccc} E & \xrightarrow{f} & \mathrm{GL}(W) \\ \downarrow & & \downarrow \\ S & \xrightarrow{\overline{f}} & Z \end{array}$$

to identify $E_{\mathrm{r}} = E/H_{\mathrm{u}} = s^* \mathrm{GL}(W)$, i.e., E_{r} is pulled back over S.

By taking a local reduction of structure group of $E_{\mathrm{r}} \longrightarrow S$ (over an open subset of S, which we can assume is S) we can reduce the structure group of $E \longrightarrow M$ over that open subset of S to a holomorphic principal H_{u}-bundle $H_{\mathrm{u}} \longrightarrow E_{\mathrm{u}} \longrightarrow M$. By definition of an infinitesimally algebraic homogeneous space, H_{u} has a complex Lie group morphism $H_{\mathrm{u}} \longrightarrow \overline{H}_{\mathrm{u}}$ to a unipotent complex linear algebraic group, injective on Lie algebras. Replacing M by a finite unramified covering space, we can assume that $\overline{H}_{\mathrm{u}}$ is connected. Let $\overline{E}_{\mathrm{u}} := E_{\mathrm{u}} \times^{H_{\mathrm{u}}} \overline{H}_{\mathrm{u}}$. The Atiyah classes of the bundles E_{u} and $\overline{E}_{\mathrm{u}}$ are identified, as the Lie algebras of the groups are identified. On a complex torus, any holomorphic principal bundle with connected unipotent complex linear algebraic structure group has a unique holomorphic flat connection by Lemma 12.1. Such a connection is associated to a representation of the fundamental group, and so varies holomorphically. □

References

[1] Alekseevsky D.V., Flag manifolds, *Zb. Rad. Mat. Inst. Beograd. (N.S.)* **6** (1997), 3–35.

[2] Atiyah M.F., Complex analytic connections in fibre bundles, *Trans. Amer. Math. Soc.* **85** (1957), 181–207.

[3] Atiyah M.F., Bott R., The Yang–Mills equations over Riemann surfaces, *Philos. Trans. Roy. Soc. London Ser. A* **308** (1983), 523–615.

[4] Biswas I., Dumitrescu S., Holomorphic Cartan geometries on complex tori, *C. R. Math. Acad. Sci. Paris* **356** (2018), 316–321, arXiv:1710.05874.

[5] Biswas I., Gómez T.L., Connections and Higgs fields on a principal bundle, *Ann. Global Anal. Geom.* **33** (2008), 19–46.

[6] Biswas I., McKay B., Holomorphic Cartan geometries, Calabi–Yau manifolds and rational curves, *Differential Geom. Appl.* **28** (2010), 102–106, arXiv:1009.5801.

[7] Biswas I., McKay B., Holomorphic Cartan geometries and rational curves, *Complex Manifolds* **3** (2016), 145–168, arXiv:1005.1472.

[8] Bogomolov F.A., On the decomposition of Kähler manifolds with a trivial canonical class, *Math. USSR Sb.* **22** (1974), 580–583.

[9] Borel A., Linear algebraic groups, 2nd ed., *Grad. Texts Math.*, Vol. 126, Springer, New York, 1991.

[10] Campana F., Connexité rationnelle des variétés de Fano, *Ann. Sci. École Norm. Sup. (4)* **25** (1992), 539–545.

[11] Campana F., Claudon B., Eyssidieux P., Représentations linéaires des groupes kählériens: factorisations et conjecture de Shafarevich linéaire, *Compos. Math.* **151** (2015), 351–376, arXiv:1302.5016.

[12] Čap A., Slovák J., Parabolic geometries. I: Background and general theory, *Math. Surv. Monogr.*, Vol. 154, American Mathematical Society, Providence, RI, 2009.

[13] Cartan É., La méthode de repère mobile, la théorie des groupes continus, et les espaces généralisés, *Actualités Sci. Industrielles*, Vol. 194, Hermann, Paris, 1935.

[14] Cascini P., Rational curves on complex manifolds, *Milan J. Math.* **81** (2013), 291–315.

[15] Douady A., Le problème des modules pour les sous-espaces analytiques compacts d'un espace analytique donné, *Ann. Inst. Fourier (Grenoble)* **16** (1966), 1–95.

[16] Ehresmann C., Les connexions infinitésimales dans un espace fibré différentiable, in Colloque de Topologie, Georges Thone, Liège, 1951, 29–55.

[17] Fujiki A., On the structure of compact complex manifolds in $\mathcal{C}$, in Algebraic Varieties and Analytic Varieties (Tokyo, 1981), *Adv. Stud. Pure Math.*, Vol. 1, North-Holland, Amsterdam, 1983, 231–302.

[18] Goldman W.M., Locally homogeneous geometric manifolds, in Proceedings of the International Congress of Mathematicians. Vol. II, Hindustan Book Agency, New Delhi, 2010, 717–744, arXiv:1003.2759.

[19] Grauert H., Peternell T., Remmert R. (Editors), Several complex variables. VII, *Encycl. Math. Sci.*, Vol. 74, Springer, Berlin, 1994.

[20] Grauert H., Remmert R., Coherent analytic sheaves, *Grundlehren Math. Wiss.*, Vol. 265, Springer, Berlin, 1984.

[21] Grauert H., Remmert R., Theory of Stein spaces, *Class. Math.*, Springer, Berlin, 2004.

[22] Hartshorne R., Algebraic geometry, *Grad. Texts Math.*, Vol. 52, Springer, New York, 1977.

[23] Hilgert J., Neeb K.-H., Structure and geometry of Lie groups, *Springer Monogr. Math.*, Springer, New York, 2012.

[24] Hochschild G.P., Basic theory of algebraic groups and Lie algebras, *Grad. Texts Math.*, Vol. 75, Springer, New York, 1981.

[25] Hörmander L., An introduction to complex analysis in several variables, 3rd ed., *North-Holland Math. Libr.*, Vol. 7, North-Holland Publishing Co., Amsterdam, 1990.

[26] Huckleberry A.T., The classification of homogeneous surfaces, *Exposition. Math.* **4** (1986), 289–334.

[27] Huybrechts D., Lehn M., The geometry of moduli spaces of sheaves, 2nd ed., *Camb. Math. Libr.*, Cambridge University Press, Cambridge, 2010.

[28] Iitaka S., Algebraic geometry, *North-Holland Math. Libr.*, Vol. 24, Springer, New York, 1982.

[29] Jacobson N., Basic algebra. I, 2nd ed., W. H. Freeman and Company, New York, 1985.

[30] Jahnke P., Radloff I., Projective uniformization, extremal Chern classes and quaternionic Shimura curves, *Math. Ann.* **363** (2015), 753–776, arXiv:1404.2848.

[31] Katzarkov L., On the Shafarevich maps, in Algebraic Geometry—Santa Cruz 1995, *Proc. Sympos. Pure Math.*, Vol. 62, American Mathematical Society, Providence, RI, 1997, 173–216.

[32] Katzarkov L., The geometry of large fundamental groups, *J. Math. Sci. (New York)* **94** (1999), 1100–1110.

[33] Kawamata Y., Minimal models and the Kodaira dimension of algebraic fiber spaces, *J. Reine Angew. Math.* **363** (1985), 1–46.

[34] Kawamata Y., On the length of an extremal rational curve, *Invent. Math.* **105** (1991), 609–611.

[35] Kobayashi S., Differential geometry of complex vector bundles, *Publ. Math. Soc. Japan*, Vol. 15, Princeton University Press, Princeton, NJ, 1987.

[36] Kobayashi S., Ochiai T., Holomorphic projective structures on compact complex surfaces, *Math. Ann.* **249** (1980), 75–94.

[37] Kollár J., Shafarevich maps and plurigenera of algebraic varieties, *Invent. Math.* **113** (1993), 177–215.

[38] Kollár J., Rational curves on algebraic varieties, *Ergeb. Math. Grenzgeb. (3)*, Vol. 32, Springer, Berlin, 1996.

[39] Lai C.-J., Varieties fibered by good minimal models, *Math. Ann.* **350** (2011), 533–547, arXiv:0912.3012.

[40] Lazarsfeld R., Positivity in algebraic geometry. I. Classical setting: line bundles and linear series, *Ergeb. Math. Grenzgeb. (3)*, Vol. 48, Springer, Berlin, 2004.

[41] Loray F., Pereira J.V., Touzet F., Singular foliations with trivial canonical class, *Invent. Math.* **213** (2018), 1327–1380, arXiv:1107.1538.

[42] Matsushima Y., Morimoto A., Sur certains espaces fibrés holomorphes sur une variété de Stein, *Bull. Soc. Math. France* **88** (1960), 137–155.

[43] McKay B., Holomorphic Cartan geometries on uniruled surfaces, *C. R. Math. Acad. Sci. Paris* **349** (2011), 893–896, arXiv:1105.4732.

[44] McKay B., Complex homogeneous surfaces, *J. Lie Theory* **25** (2015), 579–612, arXiv:1406.2487.

[45] McKay B., The Hartogs extension problem for holomorphic parabolic and reductive geometries, *Monatsh. Math.* **181** (2016), 689–713, arXiv:1302.5796.

[46] McKay B., Holomorphic geometric structures on Kähler–Einstein manifolds, *Manuscripta Math.* **153** (2017), 1–34, arXiv:1308.1613.

[47] McKay B., An introduction to Cartan geometries, arXiv:2302.14457.

[48] Milne J.S., Algebraic groups, *Camb. Stud. Adv. Math.*, Vol. 170, Cambridge University Press, Cambridge, 2017.

[49] Mori S., Threefolds whose canonical bundles are not numerically effective, *Ann. of Math.* **116** (1982), 133–176.

[50] Mumford D., Fogarty J., Kirwan F., Geometric invariant theory, 3rd ed., *Ergeb. Math. Grenzgeb. (2)*, Vol. 34, Springer, Berlin, 1994.

[51] Onishchik A.L., Vinberg E.B. (Editors), Lie groups and Lie algebras. III. Structure of Lie groups and Lie algebras, *Encyclopaedia Math. Sci.*, Vol. 41, Springer, Berlin, 1994.

[52] Pereira J.V., Touzet F., Foliations with vanishing Chern classes, *Bull. Braz. Math. Soc. (N.S.)* **44** (2013), 731–754, arXiv:1210.5916.

[53] Procesi C., Lie groups: An approach through invariants and representations, *Universitext*, Springer, New York, 2007.

[54] Rinow W., Über Zusammenhänge zwischen der Differentialgeometrie im Großen und im Kleinen, *Math. Z.* **35** (1932), 512–528.

[55] Serre J.-P., Complex semisimple Lie algebras, Springer Monogr. Math., Springer, Berlin, 2001.

[56] Sharpe R.W., Differential geometry: Cartan's generalization of Klein's Erlangen program, *Grad. Texts Math.*, Vol. 166, Springer, New York, 1997.

[57] Sharpe R.W., An introduction to Cartan geometries, *Rend. Circ. Mat. Palermo (2) Suppl.* (2002), 61–75.

[58] Siu Y.-T., Abundance conjecture, in Geometry and analysis. No. 2, *Adv. Lect. Math. (ALM)*, Vol. 18, International Press, Somerville, MA, 2011, 271–317, arXiv:0912.0576.

[59] Springer T.A., Linear algebraic groups, 2nd ed., Mod. Birkhäuser Class., Birkhäuser, Boston, MA, 2009.

[60] Ueno K., Classification theory of algebraic varieties and compact complex spaces, *Lect. Notes Math.*, Vol. 439, Springer, Berlin, 1975.

[61] Varouchas J., Kähler spaces and proper open morphisms, *Math. Ann.* **283** (1989), 13–52.

[62] Voisin C., Hodge theory and complex algebraic geometry. I, *Camb. Stud. Adv. Math.*, Vol. 76, Cambridge University Press, Cambridge, 2007.

[63] Zuo K., Representations of fundamental groups of algebraic varieties, *Lect. Notes Math.*, Vol. 1708, Springer, Berlin, 1999.